Monetary Policy Shocks, Large Language Models and Cross-Validation

How the Fed information effect has been overstated — and understated

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Motivation and This Paper

- Monetary policy moves output, inflation and employment.
- Most rate changes are a response to the economy, which makes the exogenous part hard to isolate.

Narrative approach (Romer and Romer, 2004): regress the rate change on the Fed's information; the residual is the shock.

$$\Delta i_t = \underbrace{\Gamma(X_t)}_{\text{Fed information set}} + \underbrace{\varepsilon_t}_{\text{shock}}$$

Overstated

In-sample $\hat{\Gamma}$ overfits and the shock shrinks. We estimate it out of sample, with nested cross-validation.

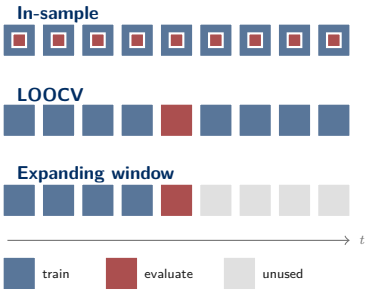
Understated

A narrow X_t leaves part of the Fed's information in the shock. We add LLM embeddings of the full Greenbook and Beige Book.

Related Literature

Strand	Key references	What we do
Monetary policy shocks	Romer and Romer (2004), Gertler and Karadi (2015), Nakamura and Steinsson (2018), Bauer and Swanson (2023), Aruoba and Drechsel (2024)	Estimate the narrative shock out of sample.
Text and the Fed's information	Handlan (2022), Acosta (2023), Aruoba and Drechsel (2024)	Use full LLM embeddings instead of a single index.
Cross-validation	Stone (1974), Cerqueira et al. (2020), Wainer and Cawley (2021)	Bring these tools to shock identification.

Cross-Validation: The Shock as an Out-of-Sample Residual



- **In sample**, every added regressor makes the shock smaller (Efron, 2004).
- **LOOCV** trains on future data, so it overstates predictability:

$$\text{MSE}_{\text{EW}} = \text{MSE}_{\text{LOOCV}} + \underbrace{\text{leakage}}_{\geq 0}$$

- The **expanding window** (EW) uses only the past, so it is the only scheme that could run in real time.

Cross-Validation in Practice: Replicating the Narrative Shocks

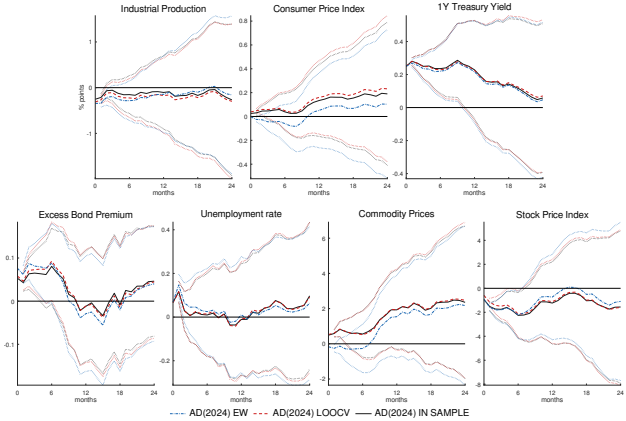
Ridge on	Regressors	In-sample R^2	EW R^2
Romer–Romer forecasts	18	0.435	0.318
Expanded forecasts	133	0.501	0.342
+ text sentiments	429	0.675	0.428
+ 4 lags	1613	**0.887**	**0.416**

210 FOMC meetings, Oct 1982 – Oct 2008.

In sample, R^2 doubles, from 0.44 to 0.89, so the shock keeps shrinking. Out of sample it only rises from 0.32 to 0.43, and adding lags does not help.

> The shock should only shrink when the out-of-sample fit improves.

Cross-Validation Changes the Impulse Responses



Shocks of Aruoba and Drechsel (2024), estimated three ways.

- The three shock series look almost identical.
- Yet the CPI and commodity-price responses switch sign under EW.

LLM Embeddings: A Broader Fed Information Set

We embed the full Greenbook and Beige Book texts with FinBERT, which beats the dictionary measure of Aruoba and Drechsel (2024) by 11% in EW R^2, and combine the embeddings with the Greenbook forecasts.

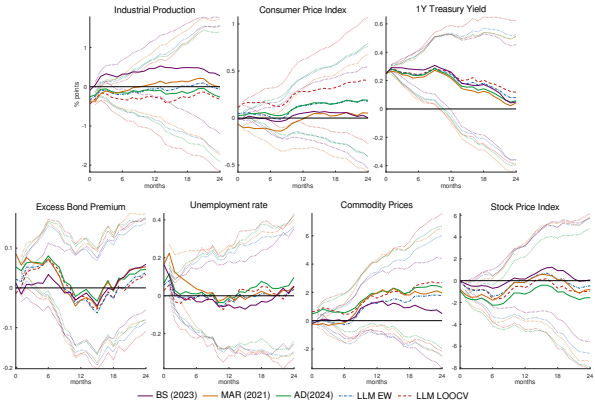
Result

XGBoost has the best LOOCV fit and ties for the best EW fit, so we use its residual as the shock. Out of sample, the estimators are close.

FinBERT embeddings + Greenbook forecasts

	In-sample R^2	EW R^2	LOOCV R^2
Ridge	0.477	0.437	0.434
Elastic Net	0.330	0.342	0.386
Random Forest	0.760	**0.450**	0.462
SVM	0.572	0.449	0.434
XGBoost	0.956	0.449	**0.501**

Final Results: Macroeconomic Effects of LLM-Based Shocks



A contractionary shock moves the economy the way theory predicts:

- Industrial production falls.
- Prices fall.

> Closer to what high-frequency identification finds.

Conclusion

- **Overstated.** Estimated in sample, the shock shrinks as information is added; the validation scheme alone can flip the CPI response.
- **Understated.** LLM embeddings of FOMC documents capture information that dictionary measures miss.
- The LLM-based shock behaves as theory predicts: the rate rises, output falls and prices fall.

Next steps

Chronologically consistent LLMs (He et al., 2025), and combining LLM information sets with high-frequency identification.

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